MAHINDRA UNIVERSITY HYDERABAD
École Centrale School of Engineering
Minor-II Examinations, October 2025

Program: B.Tech. **Branch: CM** **Year: 3rd** **Semester: I**

Subject: Financial Mathematics (MA3116)

Date: 29/10/2025 **Start Time: 02:00 PM**

Time Duration: 1.5 Hours **Max. Marks: 30**

Q 1:

Marks : 10

a) Consider a 3-period binomial model of pricing American put option. Let the initial stock price be $S_0 = 4$ per share, $u = 2$ be up factor, $d = 0.5$ be down factor, $r = 0.25$ be rate of interest per time period, $K = 5$ be strike price.

(i) Compute the optimal stopping time τ^*.

(ii) Compute the stopped option price process with respect to τ^*.

Q 2:

Marks : 10

a) Consider the scaled symmetric random walk $\{W^{(100)}(t),\, t \geq 0\}$.

(i) Determine the probability distribution of $W^{(100)}(0.3)$.

(ii) Show that $W^{(100)}(0.6) - W^{(100)}(0.4)$ is independent of $W^{(100)}(0.9) - W^{(100)}(0.6)$.

Q 3:

Marks : 10

a) Compute the stochastic integral

$$I(T) = \int_0^T W^3(s)dW(s), \quad T > 0.$$

b) Solve the following SDE using Itô-Doeblin formula

$$dX(t) = (\mu - \lambda X(t))dt + \sigma X^{1/2}(t)dW(t), \quad t \geq 0, \; X(0) = X_0.$$

Compute $E[X(t)]$, $Var[X(t)]$. Determine the probability distribution of $X(t)$.
